

Biologics (SIC 2836): every number in the results, explained

Companion to HANDOFF_BIOLOGICS.md, version 3. Numbers are from HiPerGator job 44688161 (2026-10-04), which reproduced the committed laptop run file by file. Windows: in sample = signals from 2025-01-01 to 2025-10-31; out of sample = signals from 2025-11-01 to 2026-08-31 (10 months each, the lead asked for equal halves). A trade belongs to the window of its signal date, except in the cash ledger, where each market day belongs to the window it falls in.

1. What each column means (applies to every table)

Column	What it is	How to read it
Trades	Number of trades whose signal date falls in the window.	Below about 30 the other columns are mostly noise.
Avg per trade	Mean of the per trade return. For the offering short: short the stock at the bid at the next open, buy XBI for the same dollars at the ask, close both 5 sessions later (cover at the ask, sell XBI at the bid). Return is on the dollars shorted.	The size of the effect per bet. +1% means an average bet made 1% of its stake.
Median	The middle trade.	If the median is far below the average, a few big winners carry the result.
Win rate	Share of trades with a return above zero.	A coin flip is 50%. 44% with a positive average means a few large wins against many small losses.
Sum of trade returns ("total P&L;" in the CSV)	All per trade returns added up, in percentage points.	NOT an account return. 100 trades at +1% each sums to +100% while the account might be up 20%. Use the cash ledger for account returns.
Sharpe (per trade)	Mean of the per trade return divided by its standard deviation, times $\sqrt{252 / 5}$ for a 5 day hold ($\sqrt{252 / 60}$ for the registry strategy, $\sqrt{252 / 10}$ for the team engine).	The scaling pretends you could run bets back to back all year with all your capital. It is a comparison tool across strategies, not an achievable Sharpe. 1.0 is good, 0 is nothing, below 0 loses.
95% (trades independent)	Bootstrap: resample the trades with replacement 5,000 times and recompute the Sharpe; take the 2.5th and 97.5th percentile.	The range the true Sharpe could plausibly be in, IF trades were independent. If it includes zero, the data cannot tell the strategy from nothing.
95% (signal weeks resampled)	Same, but whole signal weeks are resampled together, because trades in the same week overlap and share market shocks.	Slightly wider and more honest than the first interval. This is the lead's A5 request.
Costs: "none (quote mid to mid)"	Trade priced at the midpoint between bid and ask at entry and exit. No spread, no borrow.	The raw effect, as if trading were free.
Costs: "none (daily bars)"	Trade priced from the daily open and close instead of quotes.	Shown only to compare with the old method; same clock problem the lead flagged (A3).
Costs: "real bid/ask + 30%/yr borrow"	Sell at the real bid, cover at the real ask, XBI bought at the ask and sold at the bid (Massive quotes), minus a borrow fee of 30% a year for 5 sessions (0.60%).	The closest we can get to a real fill. The 30% borrow is an assumption; many of these stocks cannot be borrowed at all.

2. Strategy A: offering short (the main one)

Idea: a model predicts which small biotechs are about to announce a stock offering (an 8-K within 5 market days). Flagged stocks are shorted for 5 days, hedged with XBI. Model: gradient boosted trees, retrained each quarter on data at least 12 days older than the quarter, so it never sees its own test period. AUC 0.68. Of flagged company-days, 12.4% are followed by an offering against 2.0% for all eligible company-days.

Costs	Window	Trade s	Avg per trade	Median	Win rate	Sum of trade returns	Sharpe	95% (indepe ndent)	95% (weeks)
none (quote mid to mid)	in sample	88	+3.15%	+1.16%	58%	+277	1.18	-0.26 to 2.56	-0.60 to 2.84
none (quote mid to mid)	out of sample	99	+2.19%	+1.40%	56%	+217	1.23	-0.19 to 2.72	-0.38 to 2.91
none (daily bars)	in sample	88	+3.23%	+1.45%	55%	+284	1.23	-0.21 to 2.54	-0.45 to 2.82
none (daily bars)	out of sample	99	+2.43%	+1.59%	58%	+240	1.37	-0.04 to 2.88	-0.23 to 3.07
real bid/ask + 30% borrow	in sample	88	+1.03%	-0.98%	44%	+90	0.38	-1.20 to 1.78	-1.44 to 2.07
real bid/ask + 30% borrow	out of sample	99	-0.12%	-0.17%	49%	-12	-0.07	-1.48 to 1.38	-1.65 to 1.52
real costs, only entry spread under 2%	in sample	58	+2.14%	-0.54%	48%	+124	0.66	-1.28 to 2.38	-1.51 to 2.62
real costs, only entry spread under 2%	out of sample	63	+0.99%	+1.59%	56%	+62	0.56	-1.15 to 2.52	-1.19 to 2.68

Why the numbers look like this. Before costs the two halves agree: about +2 to +3% per trade, Sharpe about 1.2, but both intervals still include zero, so even the raw effect is not proven. Costs are about 1.6% per trade (bid/ask) plus 0.6% borrow, which is most of the gross, so after costs the average drops to about zero and the Sharpe to 0.38 and -0.07. The in-sample win rate after costs is 44% with a positive average: a few large wins (INMB +67%, OTLK +70%, INTS +51%) carry it; without the 5 best trades the in-sample Sharpe is -1.01. The "liquid subset" rows keep only trades whose entry spread was under 2%; costs are lower there, so the after-cost numbers are a little better, but intervals still include zero.

Matched controls (before costs, daily bars). For 83 of the 193 signals we found other eligible stocks on the same day with the same market cap, cash runway and dollar-volume bucket that the model did not flag. Those controls made -2.6% (in) and -2.9% (out) per trade; the matched picks made +6.0% and +1.7%; pick minus control +6.6% and +4.5%. Offering rate: picks 14.5%, controls 3.2%. So the picks are not just "small distressed biotech"; the model adds something. Before costs only, and not a causal claim.

Cash ledger (the account-level view, after costs)

One continuous account from the first trade to the last, every market day, idle days included. Each trade shorts 20% of the account at the entry bid and buys the same dollars of XBI at the ask; at most 5 positions; share counts are fixed at entry and converted across splits; borrow accrues daily on the short market value (30%/yr); no rebalancing; cash earns nothing. A window's result is the account's daily P&L; on the days inside it. Reconciliation: sum of trade P&L; equals the change in account value to the cent.

Window	Days	Return	Annualized	Sharpe	95% (5-day blocks)	95% (20-day blocks)	95% (40-day blocks)	Max drawdown	Win rate (taken)	Idle days
in sample	208	+3.6%	+4.4%	0.33	-1.52 to 2.31	-1.12 to 1.82	-0.99 to 1.64	-26.1%	45%	22
out of sample	207	+3.3%	+4.0%	0.30	-1.74 to 2.46	-1.52 to 2.30	-1.29 to 2.16	-35.0%	49%	12

Column	Meaning
Return	Account value at the end of the window divided by its value at the start, minus 1. This is real P&L; on real capital, unlike "sum of trade returns".
Annualized	The return scaled to a year (252 market days).
Sharpe (daily)	Mean daily return divided by its standard deviation, times sqrt(252). Comparable to Sharpe ratios people quote for funds. 0.3 is about nothing.

Column	Meaning
95% (block bootstrap)	Resample the daily returns in blocks of about 5, 20 or 40 consecutive days (keeps the clustering of good and bad stretches) 5,000 times. Longer blocks are more conservative. All include zero.
Max drawdown	Worst peak-to-trough fall of the account inside the window. -35% means at one point the account was down 35% from its high. That is the pain you would live through for a +3% result.
Idle days	Days with no position open. Cash sits there earning nothing.
Whole account	2025-01-03 to 2026-10-01: 187 trades taken (6 rejected because the entry bid was under \$1), +21.0%, Sharpe 0.47, drawdown -35.6%, borrow fees 22.5% of starting capital. By signal cohort: +\$152k from in-sample signals, -\$68k from out-of-sample signals, +\$127k from 6 September 2026 signals that fall in neither window. The out-of-sample +3.3% comes from its first two sessions (+9.0%, positions signalled in late October); after that it is -5.3%.

Conclusion for A: the model predicts offerings and the flagged stocks drop more than their peers, but trading costs and borrow eat all of it. No tradable edge is claimed. Main unknowns: real borrow fees and availability (assumed), fill sizes (quotes only), and the fact that the hold length and window split were chosen with results visible.

3. Strategy B: trial registry stage (rejected)

Idea: companies update ClinicalTrials.gov quietly. From 35 monthly AACT snapshots we detect changes: a trial starts recruiting, adds 20%+ sites, or raises its enrollment target (the company is still building the trial, so SHORT it, hedged with XBI); or enrollment is over / primary completion reached (BUY). Companies under \$2B, liquid, no 8-K on the event day or the 2 days before. Hold 60 market days. Costs: 2% round trip plus 15%/yr borrow on shorts. Per trade result is plain buy and hold over the 60 days (a short can lose more than 100%).

Costs	Window	Trade s	Avg per trade	Median	Win rate	Sum of trade returns	Sharpe	95% (independent)	95% (weeks)
none	in sample	83	+1.08%	+0.85%	51%	+90	0.05	-0.39 to 0.48	-0.41 to 0.53
none	out of sample	81	-3.58%	-2.99%	43%	-290	-0.21	-0.65 to 0.24	-0.65 to 0.24
2% round trip + 15% borrow	in sample	83	-2.74%	-2.80%	47%	-227	-0.13	-0.59 to 0.31	-0.59 to 0.33
2% round trip + 15% borrow	out of sample	81	-7.39%	-7.22%	40%	-598	-0.43	-0.92 to 0.00	-0.87 to 0.00

Why. Even before costs the effect is about zero (Sharpe 0.05) in sample and negative out of sample. The Sharpe scaling here uses $\sqrt{252/60}$, so a 60-day bet gets much less annualization than a 5-day bet; that is why the Sharpe numbers look small relative to the per trade averages. With a 2% round trip and 15% borrow over 60 days (about 3.6%), every window loses. The rules were designed on events up to June 2025, where they looked good (Sharpe 1.19 in the design half); on everything after that they failed (-1.62). The registry events also do not carry timely information: a status change is public the day it is posted, and vendors already sell alerts on it. Rejected.

4. Strategy C: buy after an offering 8-K, team engine (rejected)

Idea: when a company files an offering 8-K (tag public_offering), buy the stock at the close of the next session and hold 10 sessions. Priced by the team's shared engine (src.implementation.run_study at commit fa30fcb, unchanged) from Massive option-chain data, so only companies with traded options are priced. No trading costs in the engine.

Costs	Window	Events priced	Avg per event	Median	Win rate	Sum	Sharpe	95% (independent)	95% (weeks)
none (engine)	in sample	12	+13.42%	+5.06%	58%	+161	1.86	-0.72 to 3.95	-0.95 to 4.43
none (engine)	out of sample	21	+0.49%	-5.55%	29%	+10	0.11	-4.06 to 1.85	-3.69 to 1.58

Why. 12 and 21 events are far too few; the intervals run from about -4 to +4. The in-sample average rests on one event (IVVD rose 121% in 10 days); the out-of-sample median is -5.6% with a 29% win rate, and its average is held up by OTLK (+75%). The two windows disagree. Rejected.

5. Cross-check: the first-version offering signal through the team engine

The 155 signal days of the FIRST version of the offering model (before the leak fixes; not rerun) were given to the team engine as events and priced as a short (minus the engine's stock result), no costs. Only 7 (in sample) and 13 (out of sample) could be priced in the standard cell, because most of these companies have no traded options. Out of sample: +0.67% per event, win 46%, Sharpe 0.38, interval -4.22 to 4.17. Too few to read either way. Kept only as a record that the cross-check was done.

6. How to say it in one paragraph

We built a point-in-time dataset for 139 biologics companies and tested three strategies with the same windows and the same formulas. The trial registry and buy-after-offering strategies have no edge. The offering-short model does predict

offerings (12.4% hit rate vs 2.0% base, AUC 0.68) and its picks fall more than matched peers before costs (+6.6% and +4.5% per trade against same-day controls), with per trade Sharpe 1.18 and 1.23 in the two halves, both intervals including zero. After real bid/ask fills and an assumed 30% borrow fee the account made +3.6% and +3.3% over two 10-month windows, Sharpe 0.33 and 0.30, with drawdowns of 26% and 35%. No tradable edge is claimed. Three independent reviews found and we fixed: a split-adjusted price leak, a current-snapshot trial-data leak, a wrong market-close clock, an unclipped portfolio window, mixed price clocks in costs, a weak reproduction checker, and a comment that deleted a feature. Everything reruns on HiPerGator and matches file by file.